



UNIVERSITY OF THE PUNJAB

Eighth Semester - 2017

Examination: B.S. 4 Years Programme

Roll No.

009612

PAPER: Time Series Analysis-II

Course Code: STAT-413

TIME ALLOWED: 2 hrs. & 30 mins.

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Note: Attempt all questions.

Question No.2. Answer the following.

(4 × 5 = 20)

- a) Discuss the role of ACF & PACF in identification of a time series model.
- b) Consider the $SARIMA(1, 0, 0) \times (0, 1, 1)_{12}$ Model and show that
$$X_n(2) = X_{n-10} + \phi^2(X_n - X_{n-12}) + \theta\phi Z_{n-11} + \theta Z_{n-10}$$
- c) Describe Portmanteau Lack of fit tests..
- d) Find the MMSE forecast of X_{n+l} for an AR(1) process, with zero mean, using MA representation approach

Question No. 3. Given an AR(2) process, $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$,

(10)

Find MLE of σ_z^2 , also find approximate MLEs of ϕ_1 and ϕ_2 .

Question No. 4. Explain the meaning and importance of updating forecasts. Derive the

(10)

rule to update the forecast previously made with origin at "n" using the information X_{n+1} .

Question No. 5. For the model $(1 - B)(1 - 0.2B)X_t = (1 - 0.5B)Z_t$

(10)

Find forecast for one and two steps ahead and show that

$$X_n(l) = 1.2X_n(l-1) - 0.2X_n(l-2), \text{ for } l \geq 3$$

If $Z_n = 1$, $X_n = 4$, $X_{n-1} = 3$ & $\sigma_z^2 = 0.10$.

Also show that

$$X_n(2) = 3.64 \text{ and } \text{Var}[e_n(1)] = 0.10.$$



UNIVERSITY OF THE PUNJAB

Eight Semester 2014
Examination B.S. 4 Years Programme

Roll No.

PAPER: Time Series Analysis-II
Course Code: STAT-413

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided

Note: Attempt all Questions.

- Q.No.2. Describe the following: (20)
- Iterative procedure of finding the Least Squares estimates of and MA (1) process with non-zero means.
 - Importance of diagnostic checking stage, in Time series model building.
 - Revision of forecasts and its importance in long term forecasting.
 - Extrapolation of trend curves.
 - Akaike's Information Criterion (AIC), for model selection.

- Q.No.3. The following table gives the auto-correlation for residual in ARMA (1, 1) model fitted to a series of 100 observations. (6)

Lag	1	2	3	4	5	6	7	8	9	10
$r_k(Z_t)$	0.03	0.05	-0.01	0.12	0.06	0.04	0.11	0.10	0.08	0.05

With the help of Box-Pierce statistic check the appropriateness of the fitted model

- Q.No.4. Find the MMSE forecast of X_{n+t} for an AR (1) process, with zero mean, using MA representation approach. (4)

- Q.No.5. Given an AR (1) process (10)
- $$(1 - 0.7B)(X_t - 10) = Z_t$$
- We have $X_t: t = 1, 2, \dots, 100$ such that
- $$x_{97} = 10.6, \quad x_{98} = 10, \quad x_{100} = 9.9$$
- Forecast $x_{101}, x_{102}, x_{103}$.
- Obtain 95% confidence limits x_{101}, x_{102}
- Suppose that x_{101} turns out to be 8.8, update x_{102}, x_{103}

- Q.No.6. If $\{X_t\}$ follows an AR (1) process with the assumption that $Z_t \sim N(0, \sigma_z^2)$ (10)
- Then show that

$$\ln(L) = [\text{const} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln(1 - \phi_1^2) - \frac{1}{2\sigma_z^2} (A - 2\phi_1 B - \phi_1^2 C)]$$

Where,

$$A = \sum_{t=1}^n X_t^2; \quad B = \sum_{t=2}^n X_t X_{t-1}; \quad C = \sum_{t=2}^{n-1} X_t^2$$

Also show that MLE of σ_z^2 is:

$$\hat{\sigma}_z^2 = \frac{A - 2\phi_1 B - \phi_1^2 C}{n} \text{ And approximate MLE of } \phi_1 \text{ is } \hat{\phi}_1 = \frac{B}{C}$$



UNIVERSITY OF THE PUNJAB

Eight Semester 2015
Examination B.S. 4 Years Programme

PAPER: Time Series Analysis-II
Course Code: STAT-413

Roll No.
TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Note: Attempt all questions.

Attempt this Paper on Separate Answer Sheet provided

Q.No.2. Answer the following:

- Explain residual analysis as a tool for diagnostic checking in time series model building. (20)
- Explain why it is difficult to estimate parameters of MA process also describe the procedure of estimating MA process by using dual estimation method.
- Discuss akaike's information criteria (AIC), for model selection.
- Describe seasonal ARIMA process. Write down the model for SARIMA $(1,0,0) \times (0,1,1)_{12}$

Q.No.3. Given an AR (2) process,

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t; \text{ where } Z_t \sim N(0, \sigma_z^2)$$

Then show that

$$\ln(L) = \text{const} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2\sigma_z^2} S(\phi_1, \phi_2)$$

Where,

$$A = \sum_{t=1}^n X_t^2; B = \sum_{t=2}^n X_t X_{t-1}; C = \sum_{t=2}^{n-1} X_t^2$$

$$\text{If } \gamma_0 = \frac{(1 - \phi_2)\sigma_z^2}{(1 + \phi_2)[(1 - \phi_2)^2 - \phi_1^2]} \quad \& \quad \gamma_1 = \frac{\phi_1 \gamma_0}{(1 - \phi_2)}$$

Then evaluate M_2 and $S(\phi_1, \phi_2)$, then find MLE of σ_z^2 and suggest method of MLEs of ϕ_1 & ϕ_2 .

Q.No.4.

- Explain the meaning and importance of updating forecasts. Derive the rule to update the forecast previously made with origin at "n" using the information X_{n+1} . (2x6)
- Find the MMSE forecast of X_{n+1} for an AR(1) process, with zero mean using MA representation approach. Also find MMS forecast error for the process and discuss the structure of MMS forecast error.

Q.No.5. For the model,

$$(1 - B)(1 - 0.2B)X_t = (1 - 0.5B)Z_t$$

Find forecast for one and two steps ahead and show that

$$X_n(l) = 1.2X_n(l-1) - 0.2X_n(l-2); \text{ for } l \geq 3$$

If $\sigma_z^2 = 0.10$, also find $\text{Var}[e_n(1)]$ & $\text{Var}[e_n(2)]$ (8)



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – 2020

Paper: Time Series Analysis-II
Course Code: STAT-413 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following short questions.

(4 × 5 = 20)

- Discuss the importance of sample ACF & PACF in time series model building.
- Explain the meaning and importance of updating forecasts.
- Describe the procedure of finding the iterative least squares estimates of MA(1) process given by, $X_t = \mu + \theta Z_{t-1} + Z_t$
- If $\{X_t\}$ follows an AR(2) Process with zero mean, then find the Yule-walker estimates of the parameters of the model.

Answer the following questions.

Question No. 3. Following table gives the autocorrelations for residuals of an ARMA(1,1) model fitted to a series of 100 observations.

(08)

Lag	1	2	3	4	5	6	7	8
$r_k(Z_t)$	0.03	0.05	-0.01	0.12	0.06	-0.08	0.15	0.10

With the help of Ljung-Box-Pierce statistic, check the appropriateness of the fitted model.

Question No. 4. Given an AR(1) process, $X_t = \phi_1 X_{t-1} + Z_t$. Find MLE of σ_z^2 , also find approximate MLE of ϕ_1 .

(10)

Question No. 5. Given an AR(1) process

(12)

$$(1 - 0.7B)(X_t - 10) = Z_t$$

We have $X_t: t = 1, 2, \dots, 100$ Such that

$$X_{97} = 10.6, \quad X_{98} = 10, \quad X_{99} = 10, \quad X_{100} = 9.9 \quad \& \quad \sigma_z^2 = 0.21$$

- Forecast $X_{101}, X_{102}, X_{103}$.
- Obtain 95% confidence limits for X_{101}, X_{102} .
- Suppose that X_{101} turns out to be 8.8, update X_{102}, X_{103} .

and

$$\hat{\phi}_L = \frac{-\lambda_1^2 + \lambda_2}{1 - \lambda_1^2}$$

(2019
L-Q
3)

- Given the following information, estimate the parameters of AR(2) model - $\bar{Y} = 0.21$, $\gamma_1 = 0.87$, $\gamma_2 = 0.83$.

Solution :-

$$\phi_1 = \frac{\gamma_1 - \gamma_1 \gamma_2}{1 - \gamma_1^2} = \frac{0.87 - (0.87)(0.83)}{1 - (0.87)^2} = 0.6084$$

$$\phi_2 = \frac{\gamma_2 - \gamma_1^2}{1 - \gamma_1^2} = \frac{0.83 - (0.87)^2}{1 - (0.87)^2} = 0.3007$$

$$M = \frac{\bar{Y} - \phi_1 \bar{Y} - \phi_2 \bar{Y}}{1 - \phi_1 - \phi_2} = \frac{0.21 - (0.6084)(0.21) - (0.3007)(0.21)}{1 - 0.6084 - 0.3007}$$

$$M = 0.21$$

$$S = \sum_{t=3}^n \left\{ \right.$$

Diff w

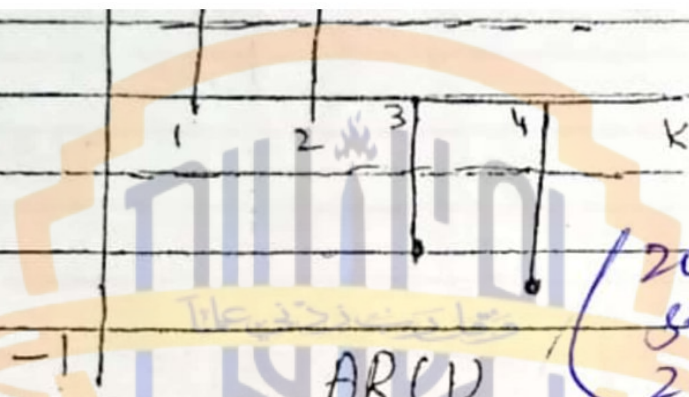
$$\frac{\partial S}{\partial \phi_1} = \frac{\partial}{\partial \phi_1} \sum_{t=3}^n Y_t$$

$$\hat{\phi}_1 \sum_{t=3}^n Y_t$$

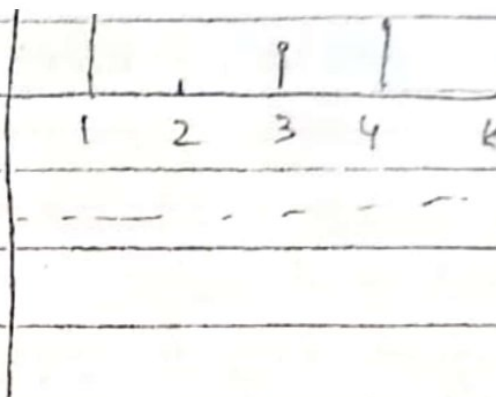
Diff w

$$\frac{\partial S}{\partial \phi_2} = \frac{\partial}{\partial \phi_2} \sum_{t=3}^n Y_t$$

6020
302
200



2017
302
200



Discuss the importance of sample ACF and PACF in time series model building -

(2018)
S-Q
2-a

(2021)
S-Q
2-a

Describe the Model-Building
131 stages suggested by Box and
Jenkins-

BOX-JENKINS MODEL BUILDING Approach

In time series analysis, the most crucial steps are to identify and build a model based on the available data. This requires a good understanding of the processes particularly. The characteristics of

$$\mu_1' = \mu_1'$$

$$\frac{a}{2} = \bar{x}$$

2015
1. Q
3. a

$$\Rightarrow \hat{a} = 2\bar{x}$$

Let $X_1, X_2, \dots, X_n$ be

Proved independent r.v's each

with the lognormal dist, find the lognormal
dist. moment estimators of μ and σ^2

Example 4:

log Normal dist is given that

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2} ; 0 < x < \infty$$

Find ME of μ and σ
OR Prove that in log Normal

• Following are the autocorrelation obtained for the residuals by fitting an ARMA (2,1) process to an observed time series with $n=200$. State the hypothesis and test the goodness of fitted model using a portmanteau test for $m=5$ till 5% level of significance.

Solution:

$$(i) H_0: \rho_k = 0, \quad H_1: \rho_k \neq 0$$

$$\begin{pmatrix} 2019 \\ LQ \\ 4 \end{pmatrix}$$

(2014 L.Q. 6)
 * Example:-

If $\{X_t\}_{t=1}^n$ follows an AR(1) process with the assumption that $Z_t \sim N(0, \sigma_z^2)$ then show that

$$\ln L = \text{Const} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} (1 - \phi^2) - \frac{1}{2\sigma_z^2} (A - 2\phi B + \phi^2 C)$$

Where $A = \sum_{t=1}^n X_t^2$, $B = \sum_{t=2}^n X_t X_{t-1}$, $C = \sum_{t=2}^{n-1} X_t^2$

Also show that MLE of σ_z^2 is $\frac{A - 2\phi B + \phi^2 C}{n}$.

Assume that the daily sale of petrol follows exponential distribution. The hypothesis H_0 , that the sale of petrol is 1000 liters per day is tested against the hypothesis that it is 1500 liters per day. If the sale on a day is 1200 liters or more, H_0 is rejected.

Find the size of type I - error, type - II error and power of the test.

Solution :-

(2015
2.9
3.6)

$\hat{\theta} = \frac{y}{n}$; $y = 0, 1, 2, \dots, n$

Example 2: Suppose $X_1, X_2, X_3, \dots, X_n$ continuous a random sample from Bernoulli dist. with
 Based on a random sample of size n from $f(x; \theta) = \theta^x (1-\theta)^{1-x}$ $x = 0, 1$
 $0 < \theta < 1$

Obtain MCSE and the MLE of θ then obtain the MCS estimator of θ .
 Compare the results.

Solution:

$f(x; \theta) = \theta^x (1-\theta)^{1-x}$; $x = 0, 1$

(2016
S.O
2.a)

Shooby described the four stages involved in time series model building?

- 1- Secular trend; which describe the movement along the term.
- 2- Seasonal variation; which represent seasonal changes.
- 3- Cyclical fluctuations; which correspond to periodical but not seasonal variations.
- 4- Irregular variations; which are other nonrandom source of variations of the series.

(2016 S-Q 2-L) If there are more than one appropriate models for time series data, on what grounds would you select the best appropriate among the candidate models.

Model Performance Measures:-

- R^2 (Degree of goodness of fit)
- $MSE = \frac{\sum e_i^2}{n}$ (Mean Square Error)
- MPE (Model Prediction Error)
- $RMSE = \sqrt{\frac{\sum e_i^2}{n}}$

Model Selection Criterion:-

- Akaike Information Criterion (AIC)
- Bayesian Information Criterion (BIC)

identified to be generated by

ss.

have an MA(1) process

(2015
5th Q
2.6)

A) Z_{t-1} Explain why it is difficult to

B) Z_t estimate parameters of MA process

Z_t also describe the procedure of

C) $Y_t = Z_t$ estimating MA process by using

$t = Z_t$ dual estimation method.

$-i = Z_t$

Example:-

(2015)
L-Q
3

Given an AR(2) process

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

Show that

$$\ln L = \text{const} - n/2 \ln \sigma_z^2 + 1/2 \ln |M_2| - 1/2 \sigma_z^2 S(\phi_1, \phi_2)$$

where $A = \sum_{t=1}^n X_t^2$ and $B = \sum_{t=1}^n X_t X_{t-1}$

$$\text{if } \gamma_0 = \frac{(1-\phi_2) \sigma_z^2}{(1+\phi_2)[(1-\phi_2)^2 - \phi_1^2]}$$

$$C = \sum_{t=2}^{n-1} X_t^2$$

$$\text{and } \gamma_1 = \frac{\phi_1}{1-\phi_2} \gamma_0$$

evaluate M_2 and $S(\phi_1, \phi_2)$

then also find MLE of σ_z^2 , and suggest method of finding MLE's of ϕ_1 and ϕ_2 .

Solution:-

Several approaches for diagnostic
re of the ... (2021) popular (2015)
e given as under (S.G. 2.6) (S.G. 2.6)

Explain residual analysis as a tool

1 Plot:- for diagnostic checking in time series

We plot the residuals and check model

any pattern still present in it. building

behaviour seems to consist of

then the fitted model is considered

one. search for some

(2020
L.Q
4)

Given AR(1) process, $X_t = \phi_1 X_{t-1} + z_t$, find MLE

EXP:- of σ_z^2 , also find approximate MLE of ϕ_1 .

Derive MLEs (approximate MLEs) for an AR(1) process given by

$$Y_t = \phi Y_{t-1} + Z_t$$

where $Z_t \stackrel{iid}{\sim} N(0, \sigma_z^2)$

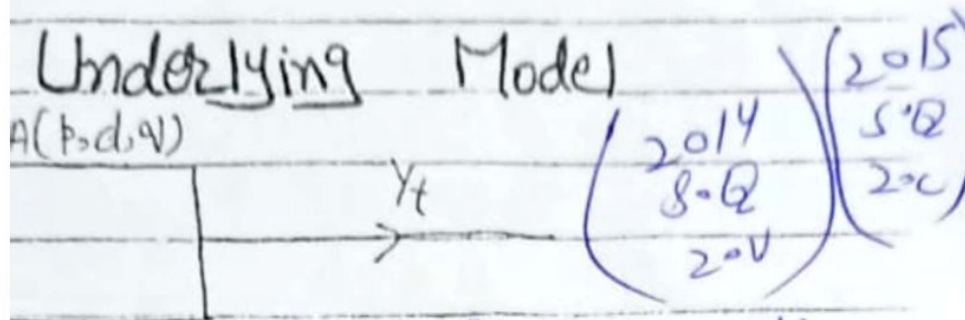
Sol:-

Suppose $\{Y_t\}_{t=1}^n$ be the observed sample time series then we can write the likelihood function as

$$l = l(y_1, y_2, \dots, y_n)$$

m. Criterion (BIC)

Information Criterion (SBIC)



$\sigma^2 = ?$ Akaike's Information

$\nabla^2 y_t$ Criterion (AIC) for
- model selection :

- AIC is a fined
- 2 technique based on in
- 4 sample fit to estimate the
- 5 likelihood of a model to
- 3 predict / estimate the future
- 2 values. A good model is the
- 4 one that has minimum AIC
- 6 among all the other models
- 4 The AIC can be used to
- 4 select between the additive
- 3 and multiplicative Holt-
- 3 winters model.

(2014
S-Q
2-iv)

(2016
S-Q
2-b)

198

Define forecasting, write a note on extrapolation of trend curves

Naive Methods:- Extrapolation involves in making statistical forecasts by using historical trends that are projected for a specified period of time into the future. It is only used for time series forecasts if we assume that past behaviour is good predictor of future behaviour, extrapolation is appealing -

t Y_t

1 Y_1

2 Y_2

3 Y_3

4 Y_4

5 Y_5

6 Y_6

7 Y_7

8 Y_8

9 Y_9

10 Y_{10}

$$= \phi \ell_{k-1}$$

$$= \hat{\phi} \ell_{k-1}$$

$$= \hat{\phi} \ell_0$$

(2020)
S.O.
20d

$$\therefore \gamma_0 = 1$$

$$= \ell_1$$

If $\{X_t\}$ follows an AR(2) process with zero mean, then find the Yule-walker estimates of the parameters of the model-

(2)

$$\phi_1 \ell_{k-1} + \phi_2 \ell_{k-2}$$

$$\hat{\phi}_1 \ell_{k-1} + \hat{\phi}_2 \ell_{k-2}$$

An AR(1) model $(1 + 0.5B)(Y_t - 50) = Z_t$ is fitted to an observed time series of 80 observations. Now a practitioner wants to use this model to forecast Y_{81} and Y_{82} . Compute the desired MMSE forecasts and 95% forecast interval if $Y_{79} = 59$, $Y_{80} = 52$, $\sigma_z^2 = 4$. Update the forecast of Y_{82} if it is observed that $Y_{81} = 62$.

Solution:

Given AR(1) process

$$(1 + 0.5B)(Y_t - 50) = Z_t$$

(2018
L.Q
3)

... the most recent observation.

pose

' $X_n(l)$ ' is the forecast of X_{n+l} with origin of forecast at $t=n$

$e_n(l)$ is the forecast error, then Show that the

$e_n(l) = X_{n+l} - X_n(l)$ MMSE forecast with origin

The MSE will be at 'n' and lead time 'l' is

$$MSE = E [X_{n+l} - X_n(l)]^2 \text{ given by}$$

$$= E [e_n(l)]^2 \quad X_n(l) = \frac{1}{\psi(B)} \left[\frac{\psi(B)}{B^l} \right] X_n$$

where $\psi(B)$ is the polynomial of the process in moving average

Suppose we have an observed time series

$\{Y_t\}_{t=1}^n$ generated by an ARMA(p,q)

If $\{Y_t\}_{t=1}^n$ follows an AR(2) process: $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$ where (Z_t) is independently and normally dist. process with 0 mean and finite variance σ_z^2 then ¹⁶⁹ (2018 L.Q 3) ²⁰²¹ (L.Q 3) show that log-likelihood function is

$$\hat{\mu} = \frac{-\bar{X} + \phi \bar{X}}{-1 + \phi}$$

$$\hat{\mu} = \frac{\bar{X}(-1 + \phi)}{(-1 + \phi)}$$

$$\hat{\mu} = \bar{X}$$

$$\ln L = \text{const} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln M_2 - \frac{1}{2\sigma_z^2} \left[\sum_{t=1}^n (Y_t - \sum_{i=1}^2 \phi_i Y_{t-i})^2 + Y' M_2 Y \right]$$

where $Y = (Y_1, Y_2) \sim N_2(0, \Sigma_2)$ and $M_2 = \sigma_z^{-2} \Sigma_2^{-1}$

* Example:-

Given an AR(2) process

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

Find the MLE of σ_z^2 and approximate MLE's of ϕ_1 and ϕ_2 .

insignificant (H_0 is accepted)

$$K \neq 0$$

we take $\max(K) = n/4$

(2017
5-Q
20C)

Describe Portmanteau
Test:- Lack of fit test. Describe

Portmanteau lack of fit test are the null
detecting underfitting is estimated hypothesis
process $\{Y_t\}$ is generated by an and use of
model. If this model is correctly portman
and fitted then following test test-

used

$$L_1 = L_2 = \dots = L_m = 0 \quad (m \geq p+q)$$

(2021
5-Q
20C)

$$\pi_i(l) = \begin{cases} \pi_i & \text{for } l=1 \\ \pi_{i+l-1} + \sum_{j=1}^{l-1} \pi_i^{(l-j)} \cdot \pi_j & \text{for } l \geq 2 \end{cases}$$

* **Question:-** Show that the forecast error for lead time 1 with different forecast origins from different origins is given by always uncorrelated

$$\text{Cov}(e_{n+l}, e_{n-s+l}) = \sigma_z^2 \sum_{i=0}^{l-1-s} \psi_i \psi_{i-s}$$

for $l > s$
, $s \neq 0$

Solution:-

As

$$e_{n+l} = \sum_{i=0}^{l-1} \psi_i z_{n+l-i}$$

• Given an AR(1) process

$$(1 - 0.7B)(X_t - 10) = z_t$$

we have $X_t, t = 1, 2, \dots, 100$ such that

$$X_{97} = 10.6, X_{98} = 10, X_{99} = 10, X_{100} = 9.9, \sigma_z^2 = 0.21$$

(a) Forecast $X_{101}, X_{102}, X_{103}$

(b) Obtain 95% confidence limits for X_{101}, X_{102}

(c) Suppose that X_{101} turns out to be 8.8, 6

update X_{102}, X_{103} - $\begin{pmatrix} 2020 \\ 1.9 \\ 5 \end{pmatrix}$

Solution :-

(a) Given AR(1) process

$$Y_t(l) = Y_t + \phi(1-\phi^l)(Y_t - Y_{t-1}) + \theta(1-\phi^l)Z_t$$

$1-\phi$ hence it is $1-\phi$ required.

Question:-

For the model

$$(1-B)(1-0.2B)X_t = (1-0.5B)Z_t$$

Find the forecast for one and two steps ahead and show that a recursive expression for forecast of 3 or more steps ahead is given by

$$X_n(l) = 1.2 X_n(l-1) - 0.2 X_n(l-2); l \geq 3$$

Find the variance of 1, 2 and 3 steps ahead

forecasts error if $Z_n=1$, $X_n=4$, $X_{n-1}=3$ and $\sigma_z^2=2$, $\sigma_z^2=0.10$

then show that $X_n(2)=3.64$ and also show that

the standard error of the corresponding $\text{Var}[e_n(1)] = 0.10$ forecast error is 1.72

If $\sigma_z^2=0.10$, also find $\text{var}[e_n(1)]$ and $\text{var}[e_n(2)]$

2020
1.6
3

Question:-

Suppose we have an observed time series $\{Y_t\}_{t=1}^{100}$ and an ARMA(2,1) is fitted to this series.

Following are the residuals autocorrelations. Apply Portmanteau test for diagnostic checking of fitted ARMA(2,1) Model. Use $m=5$

$n=100$

K	0	1	2	3	4	5	6	7	8	9	10
$\hat{\gamma}_K$	1	-0.18	-0.17	0.27	0	-0.01	-0.15	0.14	-0.09	-0.09	0.05

Solution:-

B^l

7 B.

Describe the forecast errors
t Error :- and its role in updating the
Forecast error forecasts.

$$Y_{n+1} - Y_n(l)$$
$$\sum_{i=0}^{l-1} \psi_i Z_{n+1-i} + \sum_{i=0}^{\infty} (\psi_{l+i} - \delta_i) Z_{n-i}$$

forecast, this error term

3/ Following tables gives the autocorrelations for residuals of an ARMA(1,1) model fitted to a series of 100 observations, with the help of Ljung-Box-Pierce statistic, check the appropriateness of the fitted model.

Lag	1	2	3	4	5	6	7	8
$\gamma_k(z_k)$	0.03	0.05	-0.01	0.12	0.06	-0.08	0.15	0.10

Solution:

(i) Hypotheses:

$$\begin{pmatrix} 2016 \\ L.Q \\ 4.a \end{pmatrix}$$

$$\begin{pmatrix} 2014 \\ L.Q \\ 3 \end{pmatrix} \begin{matrix} 9 & 10 \\ 0.09 & 0.05 \end{matrix}$$

Following are the autocorrelations obtained for the residuals by fitting an ARMA (1,2) process to an observed time series with $n=50$. State the null hypotheses for the goodness of fit testing model. Test the hypothesis using the Ljung-Box portmanteau test with $m=5$. What are your recommendations about the fitted model? Use 5% level of significance.

k	1	2	3	4	5	6	7	8	9	10
γ_k	-0.15	0.08	-0.05	0.01	0.03	0.01	0.05	-0.04	0.01	0.02

Solution:-

(2021
L.Q)

* Estimation Of MA Parameters

(2020)
S.G
2.6

i) Method Of least Square (Iterative)

ii) Dual Estimation

Describe the procedure of finding the iterative least squares (I) Method Of Least Square (Iterative)

Consider an MA(1) process

MA(1) process $Y_t = \mu + Z_t + \theta Z_{t-1}$

given by where μ is the mean of $\{Y_t\}$ and $\{Z_t\}$

$Z_t = \mu + \theta Z_{t-1}$ is a purely random process with mean = 0 &

finite Variance = σ_z^2

Suppose $\{Y_t\}_{t=1}^n$ be the observed time series

and we want to estimate model parameters μ

and θ . we define our objective function i.e

Sum of squared errors & given as

(2014)
S.G
2.1

(2016)
L.G
4.6

(2018)
S.G
2.6

2016
S.O.
200

221

2018
L.O.
4

Question:

Find the forecast of $\{X_{t+l}, l=1,2,3\}$ for ARIMA (1,1,1)
given by $(1-\phi B) \nabla Y_t = (1+\theta B) Z_t$

and also show that

$$\frac{Y_t(l)}{n} = Y_t + \frac{\phi(1+\phi^l)}{(1-\phi)} (Y_t - Y_{t-1}) + \frac{\theta(1-\phi^l)}{(1-\phi)} Z_t$$

Solution:-

Given process is

$$(1-\phi B) \nabla Y_t = (1+\theta B) Z_t$$

$$(1-\phi B)(1-B) Y_t = (1+\theta B) Z_t$$

$$(1-\phi B - B + \phi B^2) Y_t = (1+\theta B) Z_t$$

2020
8-Q
2-b

Explain the meaning and

Updating Forecast:-

importance of updating forecasts:- Derive the rule to update the forecast previously made

suppose we have an observed time series $\{Y_t\}_{t=1}^n$ with t we (want) have forecasted $\{Y_{n+l}; l=1,2,3\}$ with origin at forecast origin at 'n' time. Thus we forecast 'n' using the information X_n -

we assume observations at $t=n+1$ is recorded. We want to use this information i.e. Y_{n+1} to revise or update remaining forecast $\{Y_{n+l}; l=2,3, \dots\}$

(2014
L.Q
4)

(2017
S.Q
20d)

(2019
L.Q
6)

(2015
L.Q
4.b)

Question: 1

using MA approach

Find the MMSE of X_{n+1} , For an ARIMA(1) process with zero mean $\uparrow$
 $Y_t = \phi Y_{t-1} + Z_t$ Derive the MMSE forecast function for an AR(1) process.

Show that

$$Y_n(l) = \phi^l Y_n$$

$$E(e_n(l)) = 0$$

$$V(e_n(l)) = \sigma_z^2 \begin{bmatrix} 1 - \phi^{2l} \\ 1 - \phi^2 \end{bmatrix}$$

(2018
S.Q
20d)

Where $Y_n(l)$ is the MMSE forecast Y_{n+l} and $e_n(l)$ is the forecast error with origin at 'n' and lead time 'l':

An AR(1) model $Y_t - 53 = 0.65(Y_{t-1} - 53) + z_t$.

is fitted to an observed time series of 100 observations. Now, a practitioner

wants to use this model to forecast

Y_{101} and Y_{102} . (i) Compute the MMSE

forecast of Y_{101} and Y_{102} if $Y_{99} = 52.7$, $Y_{100} = 53.5$ using

(a) $\psi(B)$ polynomials

(b) $\pi(B)$ polynomials. (ii) Suppose that Y_{101}

comes out to be 51.3, update the

forecast of Y_{102} by shifting forecast

origin to $t = 101$.

Solution:

(2019)
L.Q
7

$$Y_t - 53 = 0.65(Y_{t-1} - 53) + z_t$$



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers to the following questions.

(6x5=30)

- Discuss Box and Jenkins model building stages.
- Explain Portmanteau tests.
- Define model selection criteria AIC and BIC.
- Write a short note on simple exponential smoothing.
- Discuss different forecasting procedures.
- Define Backcasting.

Write detailed answers to the following questions.

(3x10=30)

Q.2. Find the maximum likelihood estimates of ϕ_1 and ϕ_2 for the given AR(2) model.

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

Where $Z_t \sim N(0, \sigma_z^2)$

Q.3. Find the minimum mean square error forecast of X_{n+1} for AR(1) process with zero mean using MA representation. Also find its mean and variance.

Q.4. i) Suppose the Correlogram of the time series of 100 observations has

k	1	2	3	4	5	6	7	8	9	10
r_k	0.31	0.37	-0.05	0.06	-0.21	0.11	0.08	0.05	0.12	-0.01

Suggest an appropriate ARMA model. Calculate 2 sigma confidence limits for autocorrelation.

ii) A computer generates a series of 400 observations that are supposed to be random. First 10 auto-correlations of the time series are given as:

k	1	2	3	4	5	6	7	8	9	10
r_k	0.02	0.05	-0.09	0.08	-0.02	0	0.12	0.06	0.02	-0.08

Is there any evidence for non-randomness?

• An AR(1) model $(1 - 0.7B)(X_t - 50) = z_t$ is fitted to an observed time series of 100 observations. Now a practitioner wants to use this model to forecast X_{101} and X_{102} (a) Compute the desired MMSE forecasts. If $X_{99} = 49$, $X_{100} = 52$.

(b) Obtain 95% forecast limits for

X_{101} , If $\sigma_z^2 = 5$.

Solution :-

$$\begin{pmatrix} 2021 \\ L \cdot Q \\ 5 \end{pmatrix}$$

* Question:-

Consider the SARIMA $(1,0,0) \times (0,1,1)_{12}$ and
 Show that
 $X_n(2) = X_{n-10} + \phi^2(X_n - X_{n-12}) + \Theta\phi Z_{n-11} + \Theta Z_{n-10}$

Solution:-

Given Model is

SARIMA $(1,0,0) \times (0,1,1)_{12}$

i.e

$$\phi_1(B) \phi_0(B^{12}) \nabla^0 \nabla'_{12} X_t = \theta_0(B) \Theta_1(B^{12}) Z_t$$

$$(1 - \phi B) \cdot 1 \cdot 1 \cdot (1 - B^{12}) X_t = 1 \cdot (1 + \Theta B^{12}) Z_t$$

Question:-

An AR(1) model

$$(1 - 0.7B)(X_t - 90) = Z_t \quad (i) \text{ Forecast } x_{101}, x_{102}, x_{103}$$

is fitted to an observed time series of 50 obs.

Now a practitioner wants to use this model

to forecast x_{51} and x_{52} (ii) Obtain 95% confidence limit

a) Compute the required RMSE forecast if $x_{49} = 89$ of $x_{101}, x_{102}, x_{103}$

$x_{50} = 92$

b) Suppose that x_{51}^{101} comes out to be 85, update

the forecast of x_{52}^{102}, x_{103} by shifting forecast

origin to $t = 51$.

Solution:-

we have $x_t : t = 1, 2, \dots, 100$

such that $x_{98} = 10.6, x_{99} = 10, x_{100} = 9$

(2015
S-Q
20d)

Describe Seasonal ARIMA
write down the model for SARIMA

$$231 (1, 0, 0) \times (0, 1, 1)_{12}$$

SEASONAL ARIMA Model

ARIMA models are based on the assumptions
the seasonal components are deterministic
independent of other non-seasonality components.
In practice the seasonal components may
be stochastic and also correlated with
non-seasonal components. Many time series
contain seasonal periodic components.

(2010)
50 Q.
2011) In which situation the Lloyd's method is used?
525

Lloyd's method:

(Best Linear unbiased est)

Lloyd method is the
Best Linear unbiased estimates
(B.L.U.E) which discussed



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers to the following questions.

(6x5=30)

- Discuss Box and Jenkins model building stages.
- Explain Portmanteau tests.
- Define model selection criteria AIC and BIC.
- Write a short note on simple exponential smoothing.
- Discuss different forecasting procedures.
- Define Backcasting.

Write detailed answers to the following questions.

(3x10=30)

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k	1	2	3	4	5	6	7	8	9	10
r_k	0.31	0.37	-0.05	0.06	-0.21	0.11	0.08	0.05	0.12	-0.01

Suggest an appropriate ARMA model. Calculate 2 sigma confidence limits for autocorrelation.

ii) A computer generates a series of 400 observations that are supposed to be random. First 10 auto-correlations of the time series are given as:

k	1	2	3	4	5	6	7	8	9	10
r_k	0.02	0.05	-0.09	0.08	-0.02	0	0.12	0.06	0.02	-0.08

Is there any evidence for non-randomness?

(2018)

Date:

(i)

Suppose $\{Y_t\}_{t=1}^n$ be an observed time series generated by linear ^{class} stochastic process. We can further postulate, a general class of linear time series model says ARIMA(P, d, q) to build a model for observed time series.

As a time series is taken from any linear stochastic process, so we can fit any model from the general class of ARIMA model.

The three stage iterative procedure is based on.

- (i) Identification of model
- (ii) Estimation by observed time series.
- (iii) Diagnostic checking.

Identification:

In this step we identify a specific model, AR, MA, ARIMA, ARMA by studying the

(8.1.5)

Date: _____

ACF and PACF of observed time series.

Thus by identification we mean to use of data, and any other information ~~who the~~ on who the time series was generated, we also determine the order of process.

For Example:

(i) Order of MA process: The value of lag at which sample acf cut off.

(ii) Order of AR process: The value of lag at which sample pacf cut off.

Estimation:

After identification of an appropriate model or set of candidate model. we can estimate the parameters of proposed model by using several method such as ~~that is~~ maximum likelihood, least square and Yule-walker for AR process. After estimation stage, the model is particular time series.

(ii)

Consider an MA(1) process

$$Y_t = \mu + Z_t + \theta Z_{t-1}$$

where μ is the mean of $\{Y_t\}$ and $\{Z_t\}$ is a purely random process with mean $= 0$, & finite variance $= \sigma_z^2$.

Suppose $\{Y_t\}_{t=1}^n$ be the observed time series and we want to estimate model parameter μ and θ . We define our objective function i.e. Sum of Square error given as

$$S = \sum_{t=1}^n Z_t^2 = \sum_{t=1}^n (Y_t - \mu - \theta Z_{t-1})^2$$

As the above function involves unobserved series i.e. Z_{t-1} therefore we are unable to find precise estimates of μ and θ .

But we can use (same numerical) optimization technique.

(d)

$$Y_t = \phi Y_{t-1} + Z_t$$

Put $t = n+1$

$$Y_{n+1} = \phi Y_{n+1-1} + Z_{n+1}$$

1st substitution;

$$Y_{n+1} = \phi [\phi Y_{n+1-2} + Z_{n+1-1}] + Z_{n+1}$$

$$= \phi^2 Y_{n+1-2} + \phi Z_{n+1-1} + Z_{n+1}$$

2nd Substitution:

$$Y_{n+1} = \phi^2 [\phi Y_{n+1-3} + Z_{n+1-2}] + \phi Z_{n+1-1} + Z_{n+1}$$

$$= \phi^3 Y_{n+1-3} + \phi^2 Z_{n+1-2} + \phi Z_{n+1-1} + Z_{n+1}$$

(d-1) Substitution

Date: _____

$$Y_{n+l} = \varphi^l Y_{n+l-l} + \varphi^{l-1} Z_{n+l-(l-1)} + \varphi^{l-2} Z_{n+l-(l-2)} + \dots + Z_{n+l}$$

$$Y_{n+l} = \varphi^l Y_n + \varphi^{l-1} Z_{n+1} + \varphi^{l-2} Z_{n+2} + \dots + Z_{n+l} \quad (1)$$

Taking Conditional Expectation

$$Y_n(l) = \varphi^l Y(n) + \varphi^{l-1} E(Z_{n+1}) +$$

$$\varphi^{l-2} E(Z_{n+2}) + \dots + E(Z_{n+l})$$

$$Y_n(l) = \varphi^l Y(n) \quad \text{--- (2)}$$

Question No # 4.

Given process is

$$(1 - \phi B) \nabla Y_t = (1 + \theta B) Z_t$$

$$(1 - \phi B)(1 - B)Y_t = (1 + \theta B)Z_t$$

$$(1 - \phi B - B + \phi B^2)Y_t = (1 + \theta B)Z_t$$

$$Y_t - \phi Y_{t-1} - Y_{t-1} + \phi Y_{t-2} = Z_t + \theta Z_{t-1}$$

$$\text{Thus } Y_t = \frac{\phi Y_{t-1} + Y_{t-1} - \phi Y_{t-2} + Z_t + \theta Z_{t-1}}{3}$$

Put $t = t + d$

$$\text{① Thus } Y_{t+d} = \frac{\phi Y_{t+d-1} + Y_{t+d-1} - \phi Y_{t+d-2} + Z_{t+d} + \theta Z_{t+d-1}}{3}$$

put $d=1$

$$Y_{t+1} = Y_t + \phi(Y_t - Y_{t-1}) + Z_{t+1} + \theta Z_t$$

Taking conditional expectation

$$Y_{t+1}(1) = Y_t + \phi(Y_t - Y_{t-1}) + Z_t$$

put $d=2$

$$Y_{t+2} = Y_{t+1} + \phi(Y_{t+1} - Y_t) + Z_{t+2} + \theta Z_{t+1}$$

conditional expectation

$$Y_{t+2}(1) = Y_{t+1}(1) + \phi(Y_{t+1}(1) - Y_t)$$

$$y_t(2) = y_t(1) + \varphi (y_t(1) - y_t)$$

$$y_t + \varphi (y_t - y_{t-1}) + \theta z_t + \varphi (y_t + \varphi (y_t - y_{t-1}) + \theta z_t - y_t)$$

$$y_t(2) = y_t + \varphi (y_t - y_{t-1}) + \theta z_t + \varphi^2 (y_t - y_{t-1}) + \varphi \theta z_t$$

common

$$y_t(2) = y_t + \varphi (1 + \varphi) (y_t - y_{t-1}) + \theta (1 + \varphi) z_t$$

put $d=3$ in eq (1)

$$y_{t+3} = y_{t+2} + \varphi (y_{t+2} - y_{t+1}) + z_{t+3} + \theta z_{t+2}$$

Apply condition of Expectation.

$$y_t(3) = y_t(2) + \varphi (y_t(2) - y_t(1))$$

$$y_t + \varphi (1 + \varphi + \varphi^2) (y_t - y_{t-1}) + \theta (1 + \varphi + \varphi^2) z_t$$

As

$$y_t(1) = y_t + \varphi (y_t - y_{t-1}) + \theta z_t$$

$$y_t + \frac{\varphi(1-\varphi)}{(1-\varphi)} (y_t - y_{t-1}) + \theta \frac{(1-\varphi)}{(1-\varphi)} z_t$$

so

$$y_t(2) = y_t + \frac{(1-\varphi^2)}{1-\varphi} (y_t - y_{t-1}) + \theta \frac{(1-\varphi)^2}{(1-\varphi)} z_t$$

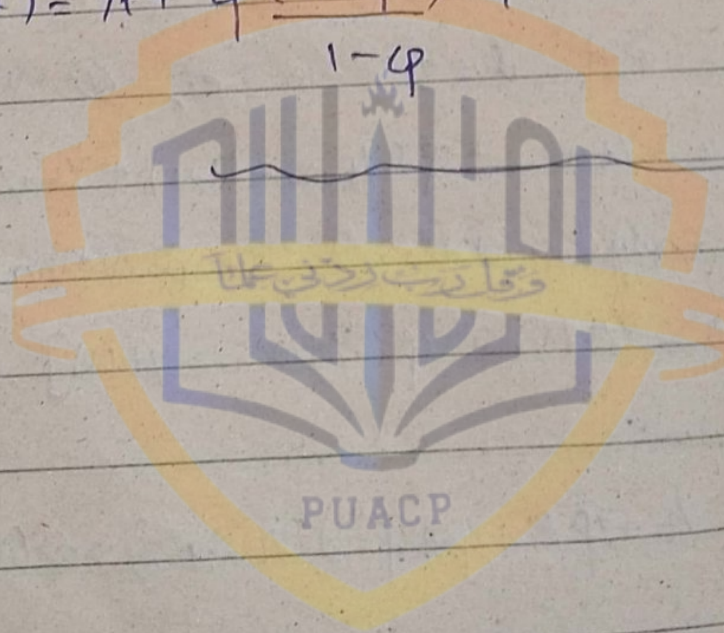
Date: _____

$$y_t(3) = y_t + \varphi(1 + \varphi + \varphi^2)(y_t - y_{t-1}) + \varphi(1 + \varphi + \varphi^2)z_t$$

$$y_t(3) = y_t + \varphi \frac{(1 - \varphi^3)}{1 - \varphi} (y_t - y_{t-1}) + \varphi \frac{(1 - \varphi^3)}{1 - \varphi} z_t$$

Thus by generalizing

$$y_t(\varphi) = y_t + \varphi \frac{(1 - \varphi^{\varphi})}{1 - \varphi} (y_t - y_{t-1}) + \varphi \frac{(1 - \varphi^{\varphi})}{1 - \varphi} z_t$$



Question No # 5

Given model is

$$(1 + 0.5B)(Y_t - S_0) = Z_t$$

$$Y_t - S_0 + 0.5Y_{t-1} - (0.5)(S_0) = Z_t$$

$$Y_t = S_0 - 0.5Y_{t-1} + (0.5)(S_0) + Z_t$$

$$Y_t = S_0 - 0.5(Y_{t-1} - S_0) + Z_t$$

Its a non zero AR(1) process
with $\mu = S_0$; $\phi = 0.5$
put $t = n+1$

$$Y_{n+1} = S_0 - 0.5(Y_n - S_0) + Z_{n+1}$$

Apply conditional Expectation

$$Y_{n+1} = S_0 - 0.5(Y_n - S_0)$$

put $t = n+2$

$$Y_{n+2} = S_0 - 0.5(Y_{n+1} - S_0) + Z_{n+2}$$

Apply conditional Expectation

$$Y_{n+2} = S_0 - 0.5(Y_{n+1} - S_0)$$

$$S_0 - 0.5[S_0 - 0.5(Y_n - S_0)]$$

$$S_0 + (0.5)^2(Y_n - S_0)$$

Date: _____

and so on

$$Y_n(d) = 50 + (0.5)^d [Y_n - 50]$$

Now $d=1$, $n=80$ $Y_{80} = 52$

$$Y_{80}(1) = 50 + (0.5)^1 [52 - 50]$$

$$Y_{80}(1) = 51$$

which is forecast of Y_{81}

put $d=2$

$$Y_{80}(2) = 50 + (0.5)^2 (52 - 50)$$

$$50.5$$

which is forecaste of Y_{82} .

PUACP

Now

Obtain 95% C.I

$$Y_{77} = 59, \quad Y_{80} = 52 \quad \sigma^2 = 4$$

$$Y_n(d) \pm Z_{\alpha/2} \sqrt{\sigma^2 \sum_{i=20}^{81} w_i^2}$$

Date: _____

we need to get weights

$$w_i = \varphi^i$$

$$w_0 = (0.5)^0 = 1$$

$$w_1 = (0.5)^1 = 0.5$$

$$w_2 = (0.5)^2 = 0.25$$

$$w_3 = (0.5)^3 = 0.125$$

95% C.I for γ_{79}

$$\gamma_{79} \pm Z_{\alpha/2} \sqrt{\hat{\sigma}^2 w_0^2}$$

$$59 \pm 1.96 \sqrt{(4)(1)}$$

$$59 \pm 1.96 \sqrt{(4)}$$

$$59 \pm 3.92$$

$$62.92, 55.08$$

95% C.I for γ_{80}

$$\gamma_{80} \pm Z_{\alpha/2} \sqrt{\hat{\sigma}^2 (w_0^2 + w_1^2)}$$

$$52 \pm 1.96 \sqrt{4(1)^2 + (0.5)^2}$$

$$52 \pm 1.96 \sqrt{(4)(1.25)}$$

$$52 \pm 1.96 (1)$$

Date: _____

~~Now~~ $52 + 1.96$

$53.96, \alpha$

50.04

Now

Update the forecast of y_{82} ,

$y_{81} = 62$

$$y_{n+1}(d) = y_n(d+1) + \alpha d (y_{n+1} - y_n(d))$$

$$y_{80+1}(d) = y_{80}(d+1) + \alpha d (y_{80+1} - y_{80}(d))$$

$$y_{80+1}(d) = y_{80}(d+1) + \alpha d (y_{81} - y_{80}(d))$$

put $d=1$

$$y_{81}(1) = y_{80}(2) + \alpha (1) (y_{81} - y_{80}(1))$$

$$50.5 + (0.5) (62 - 51)$$

$$y_{81}(1) = 56$$

which is the update forecast
of y_{82} .

Question No # 3.

Solution:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + z_t$$

let $y_1, y_2, \dots, y_n$ be a random sample of 'n' observation then likelihood function.

~~$$L = f(y_1, y_2, \dots, y_n)$$~~

$$L = f(y_1, y_2, \dots, y_n)$$

where $z_t \sim N(0, \sigma_z^2)$

$$y_t \sim N(0, \sigma_y^2)$$

We have

$$L = f(y_1, y_2) \cdot f(y_3/y_2, y_1) \cdot \dots \cdot f(y_n/y_{n-2}, y_{n-1})$$

$$L = f(y_1, y_2) \prod_{t=3}^n f(y_t / y_{t-1}, y_{t-2})$$

Taking $\ln$ on b/s

$$\ln L = \ln f(y_1, y_2) + \sum_{t=3}^n \ln f(y_t / y_{t-1}, y_{t-2})$$

①

Date: _____

As AR(2) process is

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + z_t$$

$$E[y_t / y_{t-1}, y_{t-2}] = \phi_1 y_{t-1} + \phi_2 y_{t-2}$$

$$V[y_t / y_{t-1}, y_{t-2}] = \sigma_z^2$$

So

$$(y_t / y_{t-1}, y_{t-2}) \sim N(\phi_1 y_{t-1} + \phi_2 y_{t-2}, \sigma_z^2)$$

$$f(y_t / y_{t-1}, y_{t-2}) = \frac{1}{\sqrt{2\pi\sigma_z^2}} \exp\left[-\frac{1}{2\sigma_z^2} (y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2})^2\right]$$

Taking $\ln$ on b/s

$$\ln f(y_t / y_{t-1}, y_{t-2}) = \text{const} - \frac{\ln \sigma_z^2}{2} - \frac{1}{2\sigma_z^2} (y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2})^2$$

applying summation

$$\sum_{t=3}^n \ln f(y_t / y_{t-1}, y_{t-2}) = \text{const} - \frac{n-2}{2} \ln \sigma_2^2$$

$$- \frac{1}{2\sigma_2^2} \sum_{t=3}^n (y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2})^2$$

Now dist of (y_1, y_2) is

$$f(y_1, y_2) = \frac{1}{(2\pi)^{2/2} (|\Sigma_1|)^{1/2}} \exp \left[-\frac{1}{2} (x-0)' \Sigma_1^{-1} (x-0) \right]$$

$$f(y_1, y_2) = \frac{1}{2\pi |\Sigma_2|^{1/2}} \exp \left[-\frac{1}{2} x' \Sigma_2^{-1} x \right]$$

Taking log on both sides

$$\ln f(y_1, y_2) = \text{const} - \frac{1}{2} \ln |\Sigma_2| - \frac{1}{2} (x' \Sigma_2^{-1} x)$$

As we have

$$\Sigma_2 = \sigma_2^2 M_2^{-1}$$

$$|\Sigma_2| = |\sigma_2^2 M_2^{-1}|$$

$$(\sigma_2^2)^2 |M_2|^{-1}$$

$$\ln f(x_1, x_2) = \text{const} - \frac{1}{2} \ln (\sigma_2^2)^2 |M_2|^{-1} - \frac{1}{2} (x' \Sigma_2^{-1} x)$$

$$\begin{aligned}
 &= \text{const} - \ln \sigma_2^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2} (x' \Sigma_2^{-1} x) \\
 &= \text{const} - \ln \sigma_2^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2} (x' (\sigma_2^2 M_2^{-1}) x) \\
 &= \text{const} - \ln \sigma_2^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2 \sigma_2^2} (x' M_2 x)
 \end{aligned}$$

③

Putting eq ② and ③ in eq ①

$$\begin{aligned}
 \ln L &= \text{const} - \frac{n}{2} \ln \sigma_2^2 + \ln \sigma_2^2 - \frac{1}{2 \sigma_2^2} \sum_{t=3}^n (y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2}) \\
 &\quad + \text{const} - \ln \sigma_2^2 + \frac{1}{2} \ln \sigma_2^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2 \sigma_2^2} (y' M_2 y)
 \end{aligned}$$

$$\ln L = \text{const} - \frac{n}{2} \ln \sigma_2^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2 \sigma_2^2} \left[\sum_{t=3}^n (y_t - \phi_1 y_{t-1} + \phi_2 y_{t-2})^2 + y' M_2 y \right]$$

$$\begin{aligned}
 &= \text{const} - \frac{n}{2} \ln \sigma_2^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2 \sigma_2^2} \left[\left(\sum_{t=3}^n (y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2})^2 \right) + y' M_2 y \right]
 \end{aligned}$$

Hence proved.